LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Daily Practice: Ch 13 Our Environment

Chapter / Topic: Our Environment · Week 4 of the 90-day plan

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. One day at a time, about 20 minutes each: Day 1 on the day you study the chapter, then Day 2 and Day 3. Write answers in your notebook and check them with the key the same day. Copy every mistake into your error log.
2. For every energy question, write the food chain first and the energy under each level. Each step keeps only 10%.

DAY 1: ECOSYSTEMS AND FOOD CHAINS

Q1. Which of these is an abiotic component of an ecosystem?

(a) temperature (b) bacteria
(c) grass (d) fungi

Q2. An animal that eats both plants and animals is called:

(a) a herbivore (b) a carnivore
(c) a producer (d) an omnivore

Q3. In the food chain plants → insects → frogs → snakes, the frog is a:

(a) secondary consumer (b) primary consumer
(c) tertiary consumer (d) producer

Q4. Arrange hawk, grass, snake and rat into a food chain. Which trophic level is the snake at?

Q5. Give one example each of a land food chain and a water food chain, with three trophic levels.

Q6. In a pond, you find water plants, tadpoles that eat the plants, small fish that eat the tadpoles, and bacteria on the dead remains at the bottom. Name the producer, the primary consumer, the secondary consumer and the decomposer.

DAY 2: ENERGY FLOW AND NUTRIENT CYCLING

Q7. At each trophic level, most of the energy taken in is:

(a) passed on to the next level
(b) used for life processes and lost as heat
(c) stored by decomposers
(d) reflected back to the sun

Q8. In the food chain grass → deer → lion, the grass has 20,000 J of energy. About how much reaches the lion?

(a) 20 J (b) 200 J
(c) 2,000 J (d) 20,000 J

Q9. Energy enters an ecosystem through the:

(a) decomposers (b) herbivores
(c) carnivores (d) producers (green plants)

Q10. Why does the food chain grass → human make more energy available to humans than grass → goat → human?

Q11. In the food chain phytoplankton → zooplankton → fish → kingfisher, the kingfisher receives 2 J of energy. How much energy did the phytoplankton trap? Show the energy at each level.

Q12. Energy flows through an ecosystem in one direction, but nutrients go round in a cycle. Explain the difference.

DAY 3: OZONE, WASTE AND CHALLENGE

Q13. **Assertion (A):** At ground level, ozone is a deadly poison.
Reason (R): An ozone molecule is made up of two oxygen atoms.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Q14. In 1987, an international agreement led by the United Nations Environment Programme froze the production of:

(a) carbon dioxide
(b) plastics
(c) CFCs
(d) pesticides

Q15. Which of these will take the longest to break down in soil?

(a) a banana peel
(b) a cotton cloth
(c) a newspaper
(d) a polythene bag

Q16. A school's eco-club weighed the canteen waste for one week: 40 kg of leftover food and peels, 12 kg of paper plates and napkins, 8 kg of plastic wrappers and bottles, and 2 kg of aluminium foil.

(i) Which of these wastes are biodegradable?

(ii) What percentage of the total waste is biodegradable?

(iii) Suggest one way to manage the food waste and one way to cut down the plastic waste.

Q17. Why do we never need to clean ponds or lakes, while an aquarium must be cleaned regularly?

~ Week 4 done after this one. That's all of biology. Great going! ~